2025 AMC 10A Problem 20 - Solution

Review the full statement and step-by-step solution for 2025 AMC 10A Problem 20. Great practice for AMC 10, AMC 12, AIME, and other math contests

2025 AMC 10A Problem 20

Problem:
A silo (right circular cylinder) with diameter 20 meters stands in a field. MacDonald is located 20 meters west and 15 meters south of the center of the silo. McGregor is located 20 meters east and $g > 0$ meters south of the center of the silo. The line of sight between MacDonald and McGregor is tangent to the silo. The value of g can be written as $\frac{a\sqrt{b}-c}{d}$, where a, b, c, d are positive integers, b is not divisible by the square of any prime, and d is relatively prime to $\gcd(a, c)$. What is $a + b + c + d$?

Answer Choices:

A. 119

B. 120

C. 121

D. 122

E. 123

Solution:

Let the silo's center be at the origin $O(0, 0)$, and its radius be $R = 10$.

MacDonald is located at $M(-20, -15)$, and McGregor is located at $N(20, -g)$ with $g > 0$.

The slope of their line of sight is

$$m = \frac{-g - (-15)}{20 - (-20)} = \frac{15 - g}{40}.$$

Equation of the line through $(-20, -15)$ is $y + 15 = m(x + 20)$, which simplifies to $-mx + y + (15 - 20m) = 0$.

Since the line is tangent to the circle $x^2 + y^2 = 100$, the perpendicular distance from the origin to the line equals the radius:

$$\frac{|15 - 20m|}{\sqrt{m^2 + 1}} = 10.$$

Squaring both sides:

$$\begin{aligned} (15 - 20m)^2 &= 100(m^2 + 1) \\ \implies 225 - 600m + 400m^2 &= 100m^2 + 100 \\ \implies 12m^2 - 24m + 5 &= 0. \\ \implies m &= 1 \pm \frac{\sqrt{21}}{6}. \end{aligned}$$

Since $g = 15 - 40m > 0$, we must have $m < \frac{15}{40}$, so we choose $m = 1 - \frac{\sqrt{21}}{6}$.

Therefore,

$$\begin{aligned} g &= 15 - 40\left(1 - \frac{\sqrt{21}}{6}\right) \\ &= -25 + \frac{20}{3}\sqrt{21} \\ &= \frac{20\sqrt{21} - 75}{3}. \end{aligned}$$

Thus, $a = 20, \quad b = 21, \quad c = 75, \quad d = 3$, and

$$a + b + c + d = \boxed{\textbf{(A) } 119}$$